CYBERAWARE-MIDTERM REPORT

A Web-Based Cybersecurity Awareness and Training Platform

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Course: CSIS-4495-002 (CRN 16922)
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Midterm Video Demonstration Link:
Midterm.mp4

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1. Introduction

Cybersecurity threats are becoming increasingly common in everyday digital life. From phishing emails to weak passwords and unsafe browsing habits, many users unknowingly expose themselves to security risks. While awareness resources exist, they are often passive and not interactive. This creates a gap between understanding cybersecurity theory and applying it in real-world situations.

The purpose of this project is to design and implement **CyberAware**, a web-based cybersecurity awareness platform that combines education with secure application development. The system allows users to create accounts, log in securely, complete learning modules, and track their progress.

The main research questions guiding this project are:

- How can cybersecurity awareness be delivered through an interactive web platform?
- How can authentication and data storage be implemented securely?
- How can backend integration improve realism and scalability?

CyberAware is built using React for the frontend and Supabase for backend authentication and database services. The goal is not only to create a learning platform, but also to demonstrate secure web application development practices.

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2. Summary of Initially Proposed Project

The original proposal outlined a React-based web application that would include:

- A navigation system
- Profile creation
- Cybersecurity learning modules
- Local storage-based progress tracking

At that stage, the focus was primarily on frontend development and user interface design. Progress tracking was intended to be stored in local storage for demonstration purposes.

The project aimed to provide an educational cybersecurity tool while demonstrating structured web development.

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3. Changes to the Proposal

During implementation, several important improvements were made.

3.1 Migration from Local Storage to Supabase

Initially, user data and module progress were stored in local storage. While this method allowed quick testing, it had significant limitations:

- Data could be easily manipulated through browser tools.
- Progress did not persist across devices.
- There was no real authentication system.
- It lacked backend security mechanisms.

To improve the system, the application was upgraded to use Supabase. Supabase provides:

- Secure email and password authentication
- Session management
- Cloud database storage
- Row-Level Security (RLS)
- Backend scalability

This upgrade significantly increased the technical depth and security of the project.

3.2 Implementation of Authentication and Protected Routes

Authentication was not fully implemented in the original proposal. The updated system now includes:

- Email and password signup
- Email confirmation
- Secure login
- Session tracking
- Logout functionality
- Protected routes

A custom ProtectedRoute component ensures that users must be logged in to access certain pages such as the dashboard.

This change improves access control and aligns the project with real-world web security practices.

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4. Project Planning and Timeline

Completed (Midterm Phase)

The following components have been implemented:

- React project setup
- Routing using React Router
- Navigation bar with dynamic authentication state
- Profile creation system with avatar selection
- Supabase authentication integration
- Database table creation
- Row-Level Security configuration
- Protected dashboard route
- Basic cybersecurity modules (structure)

Remaining (Final Phase)

The following features will be completed:

- Full phishing module scoring logic
- Automatic progress saving to database
- Additional cybersecurity modules
- UI/UX improvements
- Testing and debugging
- Final documentation

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5. Implemented Features

5.1 User Authentication System

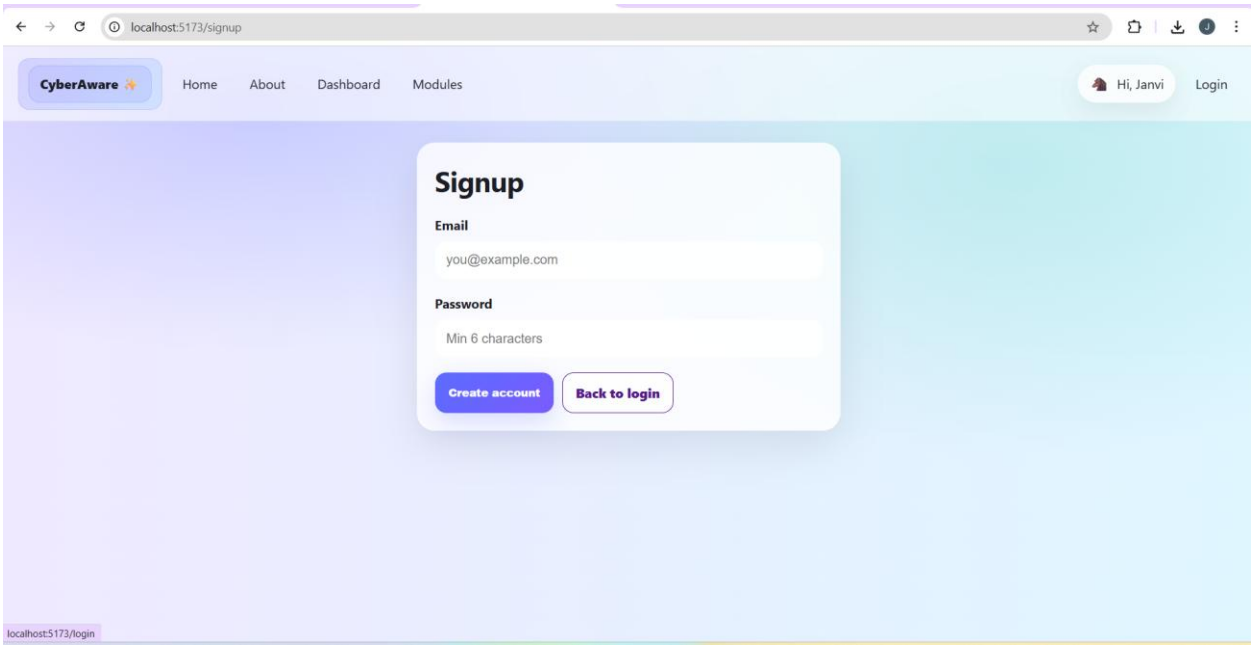
The application allows users to create an account using email and password. Upon signup, a confirmation email is sent through Supabase. After confirming the email, users can log in securely.

The system manages session state using Supabase's authentication API. The navigation bar dynamically updates based on login status.

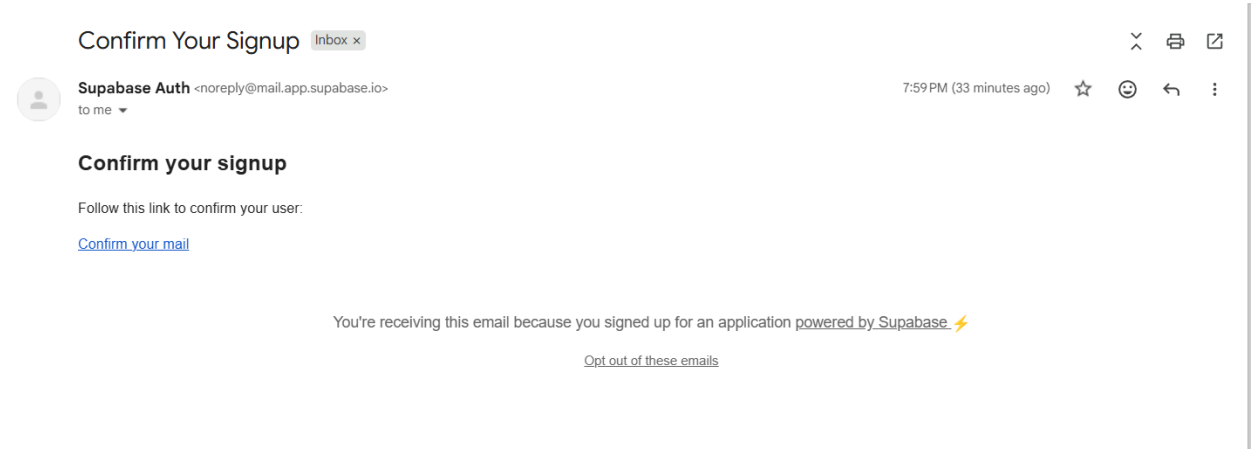
For example:

- If logged in → Logout button appears.
- If not logged in → Login button appears.

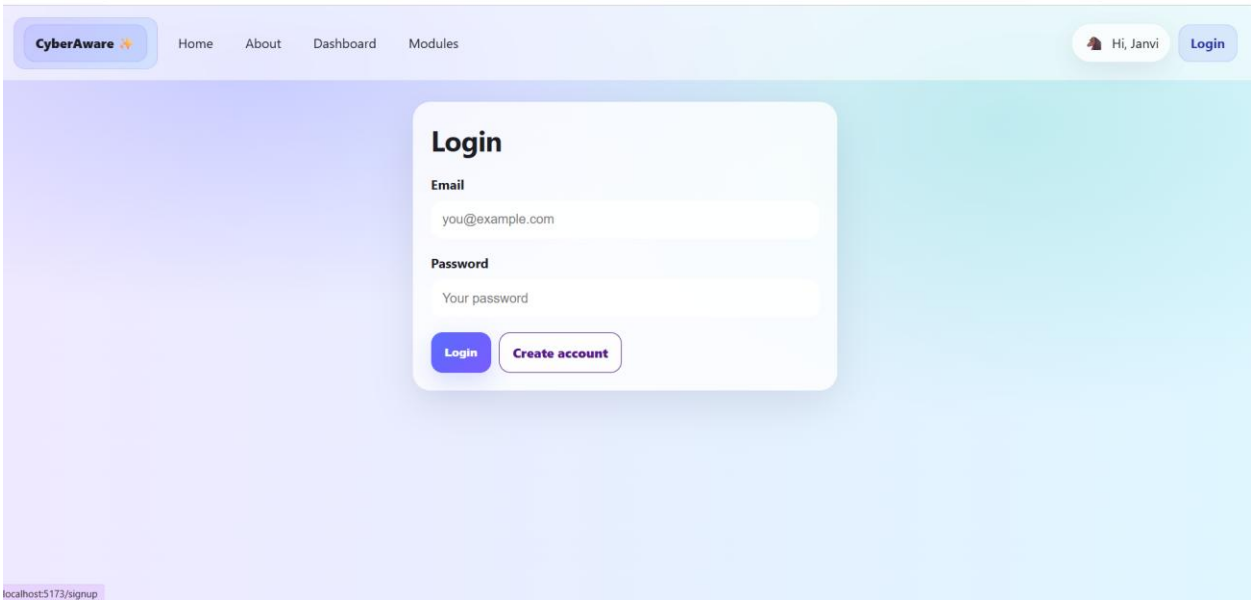
[Signup Page]



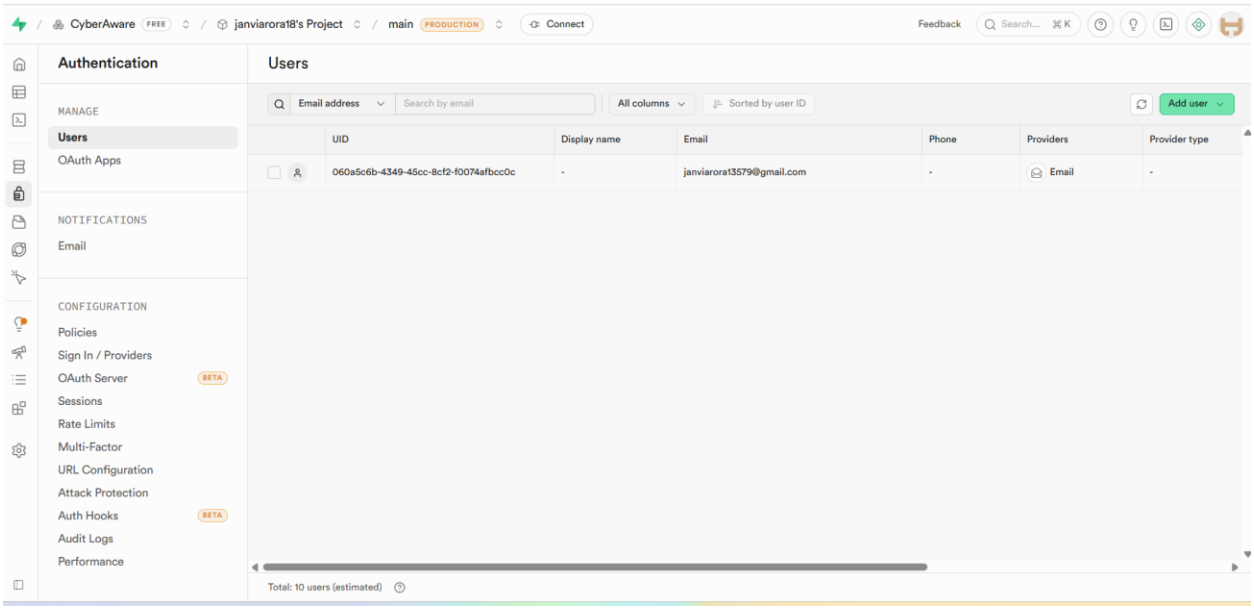
[Email Confirmation Screenshot]



[Login Page]



[Supabase Authentication Dashboard Showing User]



5.2 Protected Routes

The dashboard is protected using a custom ProtectedRoute component. If an unauthenticated user attempts to access the dashboard, they are automatically redirected to the login page.

This prevents unauthorized access to user-specific content.

5.3 Supabase Database Integration

A progress table was created in Supabase to store module completion data.

The table includes:

- user_id
- module_key
- status
- points
- created_at

Row-Level Security was enabled so that each user can only access their own records.

This ensures proper data isolation and security.

5.4 Profile Creation System

Users can create a personal profile by entering their name and selecting an avatar. The avatar and name are displayed in the navigation bar after saving.

Currently, profile information is stored locally for demonstration purposes, with plans to fully integrate profile data into the database in the final phase.

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6. AI Use Section

AI tools were used as development assistance during this project.

Tool Used: ChatGPT (GPT-5)
Account Type: Premium

AI was used for:

- Debugging routing issues
- Assisting with Supabase integration
- Structuring documentation
- Clarifying authentication errors

All coding decisions, testing, and implementation were performed manually to ensure understanding and correctness.

AI acted as a development assistant, not a replacement for implementation.

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7. Work Log

Student Name: Janvi Arora

Date	Hours	Description of Work Completed
Feb 10, 2026	2.0	Designed initial routing structure using React Router and implemented navigation links between Home, About, Dashboard, and Modules pages.
Feb 11, 2026	2.0	Built the main navigation bar with dynamic link highlighting and responsive layout styling.
Feb 12, 2026	2.5	Integrated Supabase into the React project, created project in Supabase dashboard, configured environment variables, and established secure connection.
Feb 13, 2026	2.0	Implemented email/password signup functionality using Supabase authentication API and tested account creation workflow.
Feb 14, 2026	1.5	Implemented login functionality and added session state management using Supabase auth listeners.
Feb 15, 2026	2.0	Created custom ProtectedRoute component to restrict access to dashboard for unauthenticated users.
Feb 16, 2026	2.0	Debugged routing issues related to BrowserRouter placement and NavLink context errors. Resolved blank screen issues caused by improper router configuration.
Feb 17, 2026	1.5	Created and configured Supabase progress table in SQL editor. Enabled Row-Level Security and tested user-specific access rules.
Feb 18, 2026	2.0	Implemented logout functionality and ensured navigation bar dynamically updates based on session state.

Date	Hours	Description of Work Completed
Feb 19, 2026	1.5	Refined profile creation system with avatar selection and local state management improvements.
Feb 20, 2026	2.0	Tested full authentication flow including email confirmation, login validation, and session persistence.
Feb 21, 2026	1.5	Improved UI styling for Signup and Login pages and enhanced overall visual consistency.
Feb 22, 2026	2.0	Documented Supabase database schema, RLS policies, and backend integration process for report preparation.
Feb 23, 2026	2.5	Final debugging, system testing, and preparation of midterm documentation and video script.

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8. Conclusion

The CyberAware project successfully evolved from a simple front-end application into a secure, database-integrated web platform.

The midterm implementation demonstrates:

- Secure authentication
- Session management
- Protected routing
- Backend database integration
- Row-Level Security configuration
- Structured application architecture

The next phase will focus on completing interactive learning modules and fully implementing progress tracking.

CyberAware demonstrates both technical implementation and cybersecurity awareness principles, aligning with the objectives of this course.